Research Summary: The Hamming Weight Barrier in Grover's Algorithm Under Amplitude Damping

Author: Nihar Kartikeya
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Contact: ep22btech11005@iith.ac.in

Executive Summary

This work identifies a fundamental scaling limitation in quantum search algorithms under **amplitude damping noise**. We establish the empirical law

$$P_{\text{success}} \propto e^{-pwd}$$

and demonstrate through time-resolved density matrix tomography that **state encoding designed to minimize target Hamming weight does not improve performance**.

Furthermore, we show that **Fixed-Point Amplitude Amplification (FPAA)** and **Quantum Amplitude Estimation (QAE)** variants are equally bound by this barrier, suggesting it is a **universal property of amplitude amplification algorithms that traverse Hilbert space via uniform superposition**.

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1. Core Finding: The (w ×d) Law

Under amplitude damping ((T_1) relaxation), Grover's algorithm exhibits exponential degradation scaling jointly with circuit depth (d) and target-state Hamming weight (w):

$$P_{\text{success}}(w, d) \approx e^{-pwd}$$

Symmetry Check

A comparison between targets $|000\rangle$ and $|111\rangle$ under multiple noise channels confirms that:

- **Amplitude damping** is basis-dependent
- **Phase damping** and **depolarizing noise** are Hamming-weight symmetric

Only amplitude damping penalizes high-Hamming-weight target states.

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2. Persistence of the Barrier Across Algorithms

To determine whether the barrier arises from Grover's oscillatory dynamics, we tested alternative amplitude amplification strategies.

A. Fixed-Point Amplitude Amplification (FPAA)

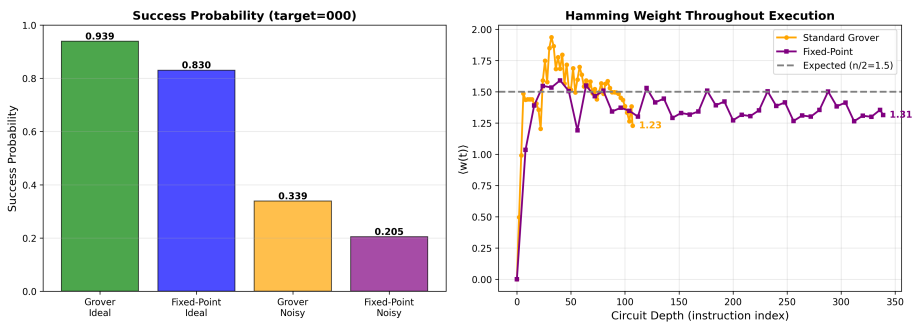

Figure 1: Fixed Point AA Results

FPAA employs a monotonic rotation schedule designed to avoid overshooting the target state.

Observations: - FPAA converges reliably in the noiseless limit - Circuit depth increases substantially
(Depth $ 80 $ vs. $ 26 $ for Grover at $ n = 3 $)

Result: - Noisy success probability drops to **20.5%**, compared to **33.9%** for Grover - The time-averaged Hamming weight still converges to

$$\langle w \rangle \approx \frac{n}{2}$$

B. Quantum Amplitude Estimation (QAE) Variants

We tested whether phase-estimation-based amplification can bypass population decay.

Result: - The Hamming-weight trajectory is statistically indistinguishable from Grover - Both methods maintain

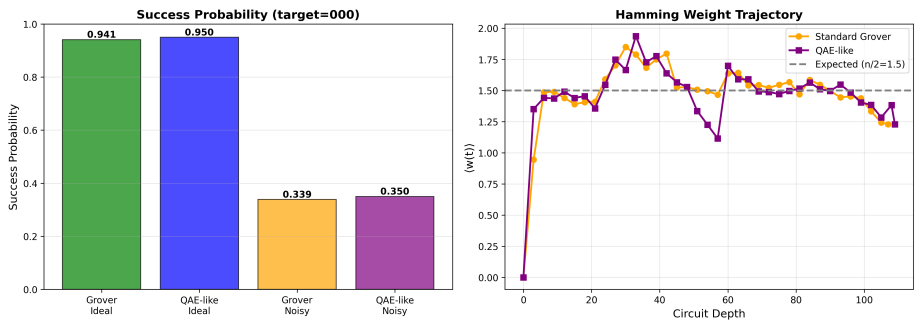

Figure 2: QAE Results

$$\langle w \rangle \approx 1.45$$

for $ n = 3 $, consistent with the expected $ n/2 = 1.5 $

Conclusion:
The barrier persists regardless of oscillatory versus fixed-point dynamics.

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3. Mechanistic Explanation: Time-Averaged Hamming Weight

Density matrix tomography reveals that the quantum state evolution is dominated by the non-target subspace:

$$|\psi(t)\rangle = \alpha(t)\,|\omega\rangle + \beta(t)\,|s^{\perp}\rangle$$

where: - $|\omega\rangle$ is the target state
- $|s^{\perp}\rangle$ spans the orthogonal uniform subspace

For an $ n $-qubit uniform superposition:

$$\langle w \rangle = \frac{n}{2}$$

Since $|\beta(t)|^2 \approx 1$ for most of the circuit execution, the system remains physically populated with excited states that are continuously exposed to T_1 decay.

Refined Decay Law

We propose that success probability depends on the **time-integrated Hamming weight**:

$$P_{\text{success}} \propto \exp\left(-p \int_0^d \langle w(t) \rangle \, dt\right)$$

which can be approximated as:

$$P_{\text{success}} \approx \exp\big(-p \langle w_{\text{avg}} \rangle d\big)$$

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4. Theoretical Implications: The Hamming Weight Barrier Conjecture

Conjecture:
Any amplitude amplification algorithm initialized in a uniform superposition must maintain a time-averaged Hamming weight

$$\langle w_{\text{avg}} \rangle \approx \frac{n}{2}$$

Why Mitigation Fails

- **Information Flow:**
  Concentrating amplitude on a target requires exploring Hilbert space, enforcing a statistically uniform population distribution.
- **No-Hiding Principle:**
  Quantum information cannot be hidden from amplitude damping without **active error correction**.

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5. Significance and Future Work

- **For NISQ Architectures:**
  Reducing **circuit depth** is more impactful than optimizing target encoding.
- **Impossibility Insight:**
  The Hamming weight barrier suggests an **intrinsic efficiency ceiling** for noisy quantum search.

- **Path Forward:**
  Only **fault-tolerant quantum error correction** can fundamentally overcome this limitation.